

The sandpile group as a coordinate system on genome reconstructions

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22 September 2026

Abstract

The reconstructions of a circular genome from its k -mer spectrum are the Eulerian circuits of its de Bruijn multigraph G , and by the BEST theorem they correspond bijectively to pairs (spanning arborescence, choice of exit order at every branch vertex). The sandpile group $\mathcal{K}(G)$ acts simply transitively on the arborescences by rotor-routing (Holroyd, Levine, Mészáros, Peres, Propp and Wilson), whose proof of BEST is itself a rotor walk. We put the two facts to a use we have not found in the literature: $\mathcal{K}(G)$ is a *torsor of coordinates on the global part of the reconstruction space*, one chip placed on a k -mer is an explicit rearrangement carrying one candidate genome to another with the same spectrum, and the difference of two candidates is a group element. We exhibit the moves, verify the torsor structure and the bijection on assembly graphs by exhaustive enumeration, and show that the classical uniform sampler of Eulerian circuits — known since Altschul and Erickson (1985) and implemented in uShuffle — is the torsor in Smith-normal-form coordinates, which is what makes the distance between two samples computable. The centrepiece is a real genome: for bacteriophage φ X174 the 12-mer spectrum leaves exactly one global choice, $\mathcal{K} = \mathbb{Z}/2$; the two reconstructions are exhibited, they are $A X B Y A Z B U$ and $A X B U A Z B Y$ for two interleaved repeated 12-mers, and a single chip exchanges the 433 and 181 bp segments. This is the $\mathbb{Z}/2$ of the interleaving theorem of the companion paper, realised in the first genome ever sequenced. Seven checks, exact except the two sampling rows.

1 Introduction

A companion paper [7] showed that the number of reconstructions of a circular sequence from its k -mer spectrum factors, by the BEST theorem, into a local part $\prod_v (d^+(v) - 1)!$ determined by branch statistics and a global part $|\mathcal{K}(G_k)|$, the order of the sandpile group, which branch statistics do not determine; and that the group structure of $\mathcal{K}(G_k)$ records repeat length, multiplicity and interleaving. This note is about the group itself rather than its order: what does the group *do* to the set of reconstructions?

What is known. Three literatures bear on the question, and we claim nothing in any of them.

- *Counting and sampling.* The BEST theorem [3, 20] counts Eulerian circuits of a directed Eulerian graph as (arborescences) $\times \prod (d - 1)!$, and since arborescences can be sampled uniformly in polynomial time [8, 18, 21], so can Eulerian circuits; Creed and Cryan state

this explicitly [9]. (For *undirected* graphs counting is #P-complete [5]; the directed case, which is ours, is easy.) Bioinformatics has used exactly this since 1985 to shuffle sequences preserving k -let counts [1, 12, 11], and to count reconstructions from k -mer spectra [2, 14].

- *Rotor-routing*. Holroyd et al. [10] prove that the sandpile group of a digraph with a sink acts freely and transitively on the spanning arborescences toward the sink (their §3), and in their §4 give a bijective proof of BEST by a rotor walk: an Eulerian tour is exactly one full turn of every rotor (Cor. 4.10). The rotor-router model itself is the “Eulerian walkers” model of Priezzhev et al. [17].
- *Sandpile groups of de Bruijn graphs*. Levine [15] computed $\mathcal{K}(\text{DB}_n)$ for the complete binary de Bruijn graph; Bidkhori–Kishore and Chan–Hollmann–Pasechnik extended this [4, 6]. None concern genome graphs.

What is here. We compose these into a statement about assembly (Theorem 1): the set of reconstructions, modulo the local exit choices, is a torsor for $\mathcal{K}(G)$. Consequences: a chip placed on a k -mer is an explicit rearrangement between two candidate genomes (Section 4); the difference of two candidates is a group element, computable as a chip word; and the classical uniform sampler is the torsor written in Smith-normal-form coordinates (Section 5). Section 6 applies this to φX174 , where the 12-mer spectrum leaves one binary choice and we show what that choice is.

What we do not claim. Not the torsor theorem, which is [10, §3]; not the bijection between tours and (arborescence, rotor order), which is [10, Cor. 4.10] and which our check (B) merely re-implements; not uniform sampling, which is [8, 18, 9] and, in bioinformatics, [1, 12, 11]. The contribution is the reading of these as a coordinate system on assemblies, the identification of chip-firing with rearrangement, and the φX174 computation.

2 Setting

S is a circular sequence, $G = G_k(S)$ its de Bruijn multigraph (vertices the distinct k -mers, one arc per $(k+1)$ -mer occurrence), which is connected and balanced with S an Eulerian circuit; $\text{Rec}(G)$ denotes the set of arc-rooted Eulerian circuits, i.e. the reconstructions [7, Rem. 1]. Fix a sink $s \in V$ and let $\text{Arb}(G)$ be the set of spanning arborescences oriented toward s ; for balanced G the count $|\text{Arb}(G)|$ and the group $\mathcal{K}(G) = \text{coker } \Delta^\top$ do not depend on s [10, Lem. 4.12]. Here $\Delta = D^+ - A$ is the out-Laplacian with the row and column of s deleted; its determinant counts arborescences *toward* s because each non-sink vertex chooses one of its out-arcs, and the transpose enters because firing v sends one chip along each out-arc, $\sigma \mapsto \sigma - \Delta^\top e_v$, so the firing lattice is the column span of $\Delta^\top$. For balanced G , $\Delta^\top = D^+ - A^\top$ is the in-Laplacian, so the choice is between two presentations of one group. A *rotor configuration* is a choice of one out-arc at every non-sink vertex; the acyclic ones are exactly $\text{Arb}(G)$. A chip placed at v is routed to the sink by the rule “advance the rotor at the current vertex by one, then move along it” [10, §3]; the map $\text{Arb}(G) \rightarrow \text{Arb}(G)$ this induces is the action of the class of e_v in $\mathcal{K}(G)$. The same action can be realised algebraically, by adding e_v to a recurrent chip configuration and toppling back to the recurrent representative, since the rotor-router group and the sandpile group coincide [10, Lem. 3.19]; we use the dynamical description because it delivers the new arborescence directly.

By the contraction lemma of [7, Lem. 4], both $\mathcal{K}(G)$ and, arc for arc, $\text{Arb}(G)$ are unchanged when every vertex of in- and out-degree 1 is smoothed away, so all of the following may be computed on the compacted (unitig) graph; the contracted arcs carry the concatenated labels and a circuit on the compacted graph is spelled back into a full-length sequence.

3 The torsor

Theorem 1 (Coordinates on reconstructions). *Let G be the de Bruijn multigraph of a circular sequence. Then*

- (i) $\mathcal{K}(G)$ acts freely and transitively on $\text{Arb}(G)$ by rotor-routing [10, Lem. 3.12, 3.13, 3.17];
- (ii) the map $(T, \rho) \mapsto$ “walk from the sink taking, at each vertex, the next unused out-arc in the cyclic order ρ_v with the tree arc of T last” is a bijection from pairs (arborescence, exit orders) onto $\text{Rec}(G)$ [10, Cor. 4.10];
- (iii) consequently $\text{Rec}(G)/\sim$, the reconstructions modulo the local exit choices, is a $\mathcal{K}(G)$ -torsor: after fixing one reconstruction, every other one is labelled by a unique group element, and $|\text{Rec}(G)| = |\mathcal{K}(G)| \cdot \prod_v (d^+(v) - 1)!$.

The theorem is a composition of two results of [10]; what we add is (iii) as a statement about genomes, and the two verifications of Section 7: check (A) enumerates $\text{Arb}(G)$ for six assembly graphs, builds the permutation representation of the chip generators, and confirms that the generated group has order $|\mathcal{K}(G)|$, a single orbit, and no fixed points; check (B) implements the walk of (ii) and confirms it is a bijection onto the brute-force enumeration of Eulerian circuits (288, 576, 144, 8 circuits on four graphs).

Remark 2 (Why the quotient). The local choices at a branch vertex of out-degree d — which of the $d - 1$ non-tree exits to take first — are what the local factor $\prod (d - 1)!$ counts, and they are independent of one another and of the arborescence. They are the ambiguity that branch statistics *do* see. The torsor lives on what remains, which [7] showed is invisible to branch statistics; Theorem 1 says it nevertheless has the structure of a finite abelian group, given explicitly by the Smith normal form of the out-Laplacian.

4 A chip is a rearrangement

Fix exit orders once and for all (say, in arc order). Then each arborescence determines one reconstruction, and adding a chip at a k -mer v moves the arborescence and hence the reconstruction. Check (M) does this on $S = \text{ACACCAAAACCA}$, $k = 2$, where $\mathcal{K}(G) = \mathbb{Z}/2 \oplus \mathbb{Z}/6$:

base reconstruction	AAAACCAACCAC	
+ one chip at CA	ACCAAAACCAAC	(6 positions changed)
+ one chip at CC	ACCAAAACCAAC	(6 positions changed)
+ one chip at AA	AAAACCAACCAC	(unchanged: $[e_{AA}] = 0$ in $\mathcal{K}$)

Each row has the 2-mer spectrum of S . The chip at AA is trivial in the group and moves nothing, which is the correct behaviour: not every k -mer carries a nonzero coordinate.

Proposition 3 (Distance). *For two reconstructions with the same exit orders there is a unique $g \in \mathcal{K}(G)$ carrying one arborescence to the other. Writing g as a sum of chip classes $[e_v]$ with*

non-negative coefficients — always possible in a finite group — gives a sequence of physically realisable moves converting the first genome into the second. The minimal number of chips $\delta(T, T')$ is a directed distance on $\text{Rec}(G)/\sim$: a quasi-metric, since $\delta(T, T')$ and $\delta(T', T)$ need not agree. Admitting chips of both signs, i.e. the symmetric generating set $\{\pm[e_v]\}$, gives the Cayley word metric of $\mathcal{K}(G)$.

This is the sense in which the group is a coordinate system rather than merely a count. Check (W) computes the word for two independently sampled reconstructions of a 27-element torsor: three chips.

One thing a chip is *not* is a single transposition. The rotor walk re-threads every branch vertex it passes on its way to the sink, so one chip can change the tree arc at several junctions at once. On the compacted φX174 graph at $k = 11$, which has ten branch vertices, the nine generators change 6, 5, 3, 2, 4, 2, 3, 1, 0 tree arcs respectively (check (J)); only one of them is a single re-threading. The directed distance of Proposition 3 is therefore not the transposition or reversal distance of comparative genomics, and we see no immediate inequality between them.

5 Sampling, and what the group adds

Uniform sampling of $\text{Rec}(G)$ is classical: sample an arborescence uniformly by a loop-erased random walk [21, 18] or by unranking [8], then choose exit orders uniformly [9]; this is what [1, 12] do and what uShuffle implements [11]. Theorem 1 gives the same distribution in a different form: with $\mathcal{K}(G) \cong \bigoplus \mathbb{Z}/d_i$ from the Smith normal form $U\Delta^\top V = D$, draw r_i uniformly mod d_i , take $U^{-1}r$ as a chip configuration, and route it from a fixed base arborescence. Check (U) draws 2700 samples this way on a torsor of order 27 and finds all 27 arborescences with $\chi^2 = 31.2$ on 26 degrees of freedom ($p = 0.22$); Wilson’s sampler on the same graph gives $\chi^2 = 22.3$ ($p = 0.67$, check (W)). The two are the same distribution. What the coordinate form adds is not speed — Wilson’s algorithm is faster — but the ability to say *where* a sample is. A shuffler such as uShuffle returns independent draws and nothing else: two of its outputs stand in no computable relation, and the sample space has no structure beyond its cardinality. In the torsor two samples differ by a computable group element, a sample can be moved by a prescribed chip, the whole space can be enumerated by stepping through Smith coordinates, and a sample can be conditioned on a prescribed value of some coordinates. What it does *not* give is a linear way to impose local constraints, as the next remark records.

Remark 4 (Local constraints are not cosets). A long read that spans a branch vertex v fixes which out-arc of v is the tree arc, selecting $S_e = \{T \in \text{Arb}(G) : T(v) = e\}$. If several such constraints could be combined by linear algebra in $\mathcal{K}(G)$, each S_e , transported to the group through the torsor, would have to be a coset of a subgroup. It is not: on the graph of ACACAAAACCA at $k = 2$ ($|\mathcal{K}| = 12$) two of the seven non-empty constraint sets are not cosets, on CCAACAACCAAC at $k = 3$ three of nine, on AABABBAABABB two of eight; only the single-repeat graph AABBAABBAABB ($\mathcal{K} = (\mathbb{Z}/3)^2$) has every constraint a coset (check (N)). The group structure is global; the local structure of arborescences is not linear in it, and a “congruence solver” for long-read disambiguation would have to be a search over the torsor, not a system of congruences.

6 φ X174: one global choice

The companion paper computed the sandpile spectrum of bacteriophage φ X174 (NC_001422.1, 5386 bp) and found $\mathcal{K}(G_{12}) = \mathbb{Z}/2$ and $\mathcal{K}(G_{11}) = \mathbb{Z}/2 \oplus \mathbb{Z}/66$ [7, Tab. 1]. Theorem 1 says that the 12-mer spectrum of the phage leaves exactly *two* global reconstructions. Check (F) exhibits them.

Proposition 5 (φ X174 at $k = 12$). *The compacted 12-mer graph of φ X174 has two branch vertices, the repeated 12-mers*

$$A = \text{CTTCTGCCGTTT}, \quad B = \text{CGTCAAGGACTG},$$

at positions 457 and 3205 (A) and 264 and 2760 (B) of NC_001422.1. Reading the reference from the second A, the genome is

$$\begin{aligned} &A X B U A Z B Y, \\ &X = 3217..263 \text{ (2433 nt, wrapping)}, \quad U = 276..456 \text{ (181 nt)}, \\ &Z = 469..2759 \text{ (2291 nt)}, \quad Y = 2772..3204 \text{ (433 nt)}, \end{aligned}$$

an interleaved pair. Its two reconstructions, both of length 5386 and both with the phage's 12-mer spectrum, are the reference $A X B U A Z B Y$ and $A X B Y A Z B U$, the segments following the two copies of B exchanged; one chip at a branch vertex carries one to the other.

The coordinates account for the whole genome: $2433 + 181 + 2291 + 433 + 4 \cdot 12 = 5386$, the four unique segments and the four occurrences of the two 12-mers; the exchange permutes segments, so both reconstructions have the same length and, segment by segment, the same content.

This is precisely the situation of the interleaving theorem [7, Thm. 6] at $s = 1$: a single repeat of multiplicity two leaves no circular ambiguity (its two threadings are rotations of one another), so a surviving $\mathbb{Z}/2$ at $k = L$ must come from two interleaved repeats, and the one bit it records is which of the two B -exits follows which A . The first genome ever sequenced has exactly one such pair of 12-mers, and the group finds it. At $k = 11$ the 132 classes of $\mathbb{Z}/2 \oplus \mathbb{Z}/66$ are the corresponding statement one level down; we have not analysed their structure.

The biology of the one bit is worth stating, because it is not what one might hope. By the feature table of NC_001422.1, the exchanged segment U (276..456) spans the junction of genes C (133..393) and D (390..848), and Y (2772..3204) spans the junction of the spike genes G (2395..2922) and H (2931..3917); the four repeat occurrences lie in C , D , G and H respectively. Exchanging a C/D junction for a G/H junction breaks the reading frames on both sides, so the alternative reconstruction is not a viable isomer of the phage. It is the misassembly that a 12-mer assembler is at risk of producing, and the group locates it exactly: one bit, two named 12-mers, two named segments.

7 Verification

The script `rotor_assembly.py` runs the eight checks of Table 1 in about five seconds, all passing; its output is archived as `rotor_assembly_output.txt`, and the φ X174 sequence is distributed with it.

key	check	mode
(A)	Theorem 1(i) on six assembly graphs, $ \mathcal{K} \leq 27$: free and transitive	exact
(B)	Theorem 1(ii): the walk is a bijection onto the enumerated circuits	exact
(M)	Section 4: chips are spectrum-preserving rearrangements	exact
(U)	Smith-coordinate sampler uniform, $\chi^2 = 31.2$ on 26 df	sampled
(W)	Wilson’s sampler agrees; the chip word between two samples	sampled
(F)	Proposition 5	exact
(J)	junctions re-threaded per chip on φ X174 at $k = 11$: 0 to 6	exact
(N)	Remark 4: constraint sets are not cosets on 3 of 4 graphs	exact

Table 1: The verification battery (`rotor_assembly.py`), 8/8.

8 Scope

Quoted. The torsor and the rotor proof of BEST [10]; uniform sampling [8, 18, 21, 9] and its bioinformatic use [1, 12, 11]; counting [2, 14]; the spectrum of φ X174 and the interleaving theorem [7].

New here, as far as we can determine. Theorem 1(iii) as a statement about assemblies; chip-firing as rearrangement and the word metric of Proposition 3; the coordinate form of the sampler; and Proposition 5.

Open. (a) *Double strands.* Real assembly graphs are bidirected [13, 16]. The reverse-complement map sends an arc $u \rightarrow v$ to $\bar{v} \rightarrow \bar{u}$, reversing direction, so the passage from the k -mer digraph to the bidirected graph is not a covering of digraphs and the covering theory of critical groups [19] does not apply as it stands; a sandpile theory for bidirected de Bruijn graphs would have to start from the signed or bidirected Laplacian [22], and we know of none. (b) The 132 classes of φ X174 at $k = 11$, and more generally what the invariant factors of a real genome’s group mean position by position. (c) Whether the directed distance of Proposition 3 relates to any edit distance between assemblies; since a single chip can re-thread many junctions (Section 4), it is not the transposition distance, and any relation would have to pass through the structure of rotor walks on the compacted graph.

Provenance

Unrefereed preprint. Theorem 1 is a composition of published results and is verified, not reproved; Proposition 3 is immediate from it; Proposition 5 is an exact computation reproducible in seconds. We searched for prior use of the sandpile torsor on genome graphs and for any critical-group treatment of bidirected de Bruijn graphs, and found neither; a search cannot establish a negative.

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